

Vaishnavi Thirumala

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PROFESSIONAL SUMMARY

Artificial Intelligence & Data Science student specializing in production-grade MLOps pipelines, Generative AI architectures, and RAG systems. Proven expertise in hybrid retrieval pipelines, agentic workflows, and automated evaluation frameworks for LLM-based applications.

EDUCATION

Indian Institute of Information Technology, Sri City | CGPA: **9.53**
Bachelor of Technology – Artificial Intelligence and Data Science

Chittoor, AP
Aug 2023 – May 2027

EXPERIENCE

Eazily Done

May 2026 – July 2026

Software Engineering Intern

Remote

- **Multi-Bank Financial RAG Architecture:** Architected a multi-bank retrieval pipeline using **LlamaParse Vision AI**, **Hierarchical Node Parsing**, and metadata-aware **ExactMatch filters** to eliminate cross-bank hallucinations across complex financial PDFs.
- **LLM Fine-Tuning & Synthetic Data Pipeline:** Fine-tuned **LLaMA-3-8B (QLoRA)** and **BGE-Large** embeddings across multi-GPU setups (**PyTorch FSDP/DDP**), leveraging a **Gemini 2.5 Flash** synthetic QA pipeline to enforce anti-hallucination guardrails.

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, R (Familiarity)

ML/AI: PyTorch, TensorFlow, Scikit-learn, NLP, Reinforcement Learning

Gen AI & LLMs: LangChain, LangGraph, LlamaIndex, RAG, Hybrid Search (BM25 + Vector), Gemini API

Data & Tools: Pandas, NumPy, SciPy, MLflow, DVC, Git, Ragas, ChromaDB

Backend/Systems: FastAPI, Docker, Streamlit

Analytics: SHAP/LIME, A/B Testing, Bayesian Inference

PROJECTS

ChurnOps: Automated MLOps Pipeline for Churn Prediction

[GitHub](#)

- Built a stacking ensemble (XGBoost, LightGBM, RF + LR meta-learner, 5-fold CV) lifting accuracy **81.2%→90.4%** and ROC-AUC **0.84→0.91** on IBM Telco (7,043 rows).
- Designed a threshold optimizer across 100 probability intervals against LTV (\$500) and campaign cost (\$50), maximizing net profit to **~\$40,800 per 1,000 customers (560%+ ROI)**.
- Deployed a self-healing retraining loop via **Evidently AI** drift detection + **GitHub Actions** to auto-trigger DVC retraining and MLflow promotion when drift exceeds 50%.
- **Tech:** *XGBoost, LightGBM, Optuna, SHAP, DVC, MLflow, Evidently AI, Prometheus, Grafana, Docker*

Multi-Hop RAG Fact Verification System

[GitHub](#)

- Built a multi-hop RAG pipeline over a Wikipedia corpus achieving **66.67% Recall@5** on FEVER benchmarks at **1.2s latency on CPU-only hardware**.
- Combined **BM25 Okapi** sparse search with in-memory **FAISS** dense reranking (*all-MiniLM-L6-v2*, 384-dim) keeping the full pipeline **<1GB RAM** with zero cloud dependency.
- Used **spaCy NER** (PERSON, ORG) for dynamic query expansion enabling multi-hop bridging across disjoint documents, fed into *cross-encoder/nli-distilroberta-base* for 3-class verdict classification.
- **Tech:** *BM25Okapi, FAISS, SentenceTransformers, spaCy, HuggingFace, FastAPI, Pyvis, NetworkX*

Personalized Email Campaign Generator with A/B Testing

[Live](#) | [GitHub](#)

- Orchestrated a stateful **LangGraph** agent across multi-LLM providers (Gemini, GPT-4o, Claude) to generate persona-conditioned email variants per CRM segment with structured JSON output.
- Built a Bayesian A/B engine using **Beta-distribution priors + Monte Carlo simulation** ($N = 100,000$) to evaluate $P(\text{CTR}_A > \text{CTR}_B)$, auto-declaring winners at **95% confidence**.
- Implemented closed-loop prompt optimization appending winning attributes to a per-segment few-shot bank, steering LLM output over up to **5 iterative optimization loops**.
- **Tech:** *LangGraph, LangChain, Gemini API, GPT-4o, Claude API, FastAPI, SciPy, MLflow, Streamlit, Docker*

POSITIONS OF RESPONSIBILITY

NLP Domain Lead

Epoch – AI/ML Club of IIIT Sri City

Aug 2025 – April 2026

Sri City, AP